

CSS411 Cryptography and Network Security

Assignment

1. In the case of an EC defined over a finite field of integer modulus 23 given by $y^2 = x^3 + 3x + 8$. Given $P = (0,10)$, $Q = (7,2)$. Calculate $P+Q$. [10 marks]

2. Check whether the following points are on elliptic curve $y^2 = x^3 + 5x + 7$ over finite field modulo 23.

(a) (1,6)

(b) (1,5)

Hint: check whether $y^2 \bmod 23 = (x^3 + 5x + 7) \bmod 23$

3. In the case of an EC defined over a finite field of integer modulus 17 given by $y^2 = x^3 + 2x + 3$. Given $P = (5,11)$. Calculate $2P$ and $3P$. [5 marks]
[10 Marks]

4. A sender wants to transmit the message:

$M = \text{"HELLO"}$

For a simple integrity check, a toy hash function is defined as:

$H(M) = (\sum \text{ASCII values of all characters in } M) \bmod 100$

The ASCII values are:

- H = 72
- E = 69
- L = 76
- L = 76
- O = 79

The sender sends the message along with its hash value.

- (a) Compute the hash value of the message "HELLO".
- (b) Suppose during transmission the message is changed to:
 $M' = \text{"HELMO"}$
- (c) Compute the new hash value.
- (d) Based on the result, determine whether the receiver can detect the modification.
- (e) Explain one limitation of this toy hash function in real-world security applications.

[10 marks]

5. Write a program using suitable cryptographic library functions to implement digital signatures using both Elliptic Curve Digital Signature Algorithm (ECDSA) and RSA Digital Signature.

- a) Generate an ECDSA public-private key pair using a standard elliptic curve such as SECP256R1.
- b) Generate an RSA public-private key pair with a key size of at least 2048 bits.
- c) Accept a message from the user.
- d) Generate a digital signature for the message using:
 - a. ECDSA with SHA-256
 - b. RSA with SHA-256 and an appropriate signature padding scheme such as RSA-PSS
- e) Display the generated signatures.
- f) Verify both signatures using the corresponding public keys.
- g) Modify the original message and demonstrate that signature verification fails for the modified message.

Use suitable library functions available in Python's cryptography library or an equivalent cryptographic library. [10 Marks]
